CT51 | A degree-eight cross-characteristic torsion bridge

SCOPE: explicit fixed-instance number-field specialization; [pcc:unverified].
Curve: y^2 + y = x^3 - x^2 - 10x - 20 (Cremona 11a1).
Global 3-division polynomial:
    f(X) = 3X^4 - 4X^3 - 60X^2 - 237X - 21.
Global point: (X,Y), Y^2+Y=X^3-X^2-10X-20.

EXACT DEGREE EIGHT
1. Mod 2, f(X) = X^4+X+1, irreducible over F2.
2. Over F16, R=X^3+X^2 has absolute trace one.
3. Y^2+Y=R is irreducible over F16, giving [K:Q]=8.
4. All nonzero 3-torsion on this curve has degree eight over F2.

TWO CHARACTERISTICS
Prime 29: f(18)=0, Y=5, P=(18,5), exact order 3.
Binary: F256=F16[Y]/(Y^2+Y+R), P'=(X,Y), exact order 3.
Subgroup bridge: O->O, (18,5)->(X,Y), (18,23)->(X,Y+1).
Source full group: 30 points. Target full group: 225 points.

EXECUTABLE CONTROL
Independent Python tower and JavaScript degree-8 polynomial field.
Both check 9/9 addition pairs, 3/3 toy DLPs, full 8 geometric
3-torsion points and public-generator orientation controls.

NEGATIVE / COMPLEXITY RESULT
No nonzero 3-torsion over F_(2^m) for m < 8 on this curve.
The pointwise inverse specialization is a three-entry lookup.
No uniform arbitrary-point algorithm or ECDLP speedup follows.

REPRODUCE: isogeny-labs/crypto PR #4015 (stacked on #4012).
SOURCES: Masser-Wuestholz 1993; Lozano-Robledo 2015;
crypto problems/lift-reduce-second-prime-transfer.md.
No paid compute; scientific PCC signing remains outstanding.